PIPELINE AUDIT AND HEALTH REPORT

Software: CI/CD pipeline

Commits (features, hotfixes, docs changes) are built on three runners that sometimes crash, then unit-tested (2% flaky failures retry the build). Docs skip integration and go straight to deploy; everything else runs in one of three shared integration environments. Hotfixes jump every queue. The nodes here are build servers, but the maths is the same as a factory.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 61 | 10 replications

HEALTH SCORE

94 /100

Grade A

Healthy: the design is robust and has headroom.

Critical 0 | High 0 | Medium 1 | Low 2 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
23.7 units/h	183 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
12.7 units	1954 s	int_env_2

Executive summary

The line keeps up with demand (0.00% lost, sources never throttled): throughput equals the arrival rate, so nothing is binding right now. The busiest station is 'int_env_2' at 84% busy; that is where capacity runs out first as demand grows.

What to fix first

- Medium (unit_tests): Speed up or add capacity downstream of this station, or enlarge the buffer after it.
- Low (throughput): Use a longer run if you need tighter estimates.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

Best measured improvement

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Flow	**unit_tests**	Blocked 25% of the time: finished parts cannot leave because the next buffer is full. Action: Speed up or add capacity downstream of this station, or enlarge the buffer after it.	-3

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Low	Confidence	**throughput**	Throughput 95% CI is +-5.3% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Low	Model	**run length**	The measured window (7.5 h) is only 14x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**int_env_2**	Highest utilisation in the line (84%); the line currently keeps up with demand. Action: This is where capacity runs out first as demand grows.	
Info	Reliability	**dependencies**	2 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
commits	source	-	-	-	-	-
runner_1	machine		78.6	12.9	2.0	1.6 / 3
runner_2	machine		53.3	9.9	1.3	1.0 / 3
runner_3	machine		27.9	7.8	0.0	0.6 / 3
unit_tests	machine		58.2	25.1	0.0	3.6 / 7
int_env_1	machine		69.7	0.0	3.4	1.3 / 4
int_env_2 bottleneck	machine		84.0	0.0	0.2	2.4 / 4
int_env_3	machine		53.7	0.0	8.0	1.3 / 4
deploy	machine		59.4	0.0	0.0	1.0 / 5
prod	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	22.51 units/h	+- 1.19	1.66	5.3%
Average WIP	13.36 units	+- 2.41	3.37	18.1%
Cycle time	2061.56 s	+- 371.26	519.02	18.0%
Arrivals lost	2.32 %	+- 2.18	3.04	93.6%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
runner_1	120 min	5.0 min	96.0%	2.0%	3
runner_2	120 min	5.0 min	96.0%	1.3%	3
runner_3	120 min	5.0 min	96.0%	0.0%	0
int_env_1	240 min	15.0 min	94.1%	3.4%	3
int_env_2	240 min	15.0 min	94.1%	0.2%	1
int_env_3	240 min	15.0 min	94.1%	8.0%	1

Single points of failure

All units must pass through: unit_tests, deploy. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 185 = completed 183 + scrapped 0 + lost 0 + in system 2
Little's Law	PASS	L = 12.73, lambda x W = 12.88 (gap 1.2%)
Run length	WARN	window = 14 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics

CHECK	RESULT	DETAIL
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 23.5 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
int_env_1	10% faster processing	23.6	**+0.1**	+0.4%
runner_1	+2 buffer slots	23.5	**+0.0**	+0.0%
int_env_1	20% shorter repairs	23.5	**+0.0**	+0.0%
int_env_2	10% faster processing	23.5	**-0.0**	-0.2%
int_env_2	20% shorter repairs	23.5	**-0.0**	-0.2%
int_env_1	+2 buffer slots	23.5	**-0.0**	-0.2%
int_env_2	+2 buffer slots	23.4	**-0.1**	-0.4%
runner_1	10% faster processing	23.2	**-0.3**	-1.3%
runner_1	20% shorter repairs	23.2	**-0.3**	-1.3%

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
commits	source	interarrival=exponential(150), mix={"docs":0.2,"feature":0.7,"hotfix":0.1}, priority={"docs":0,"feature":1,"hotfix":3}
runner_1	machine	capacity=4, discipline=priority, mtbf=7200, mttr=300, on_failure=restart, process=normal(240,60)
runner_2	machine	capacity=4, discipline=priority, mtbf=7200, mttr=300, on_failure=restart, process=normal(240,60)
runner_3	machine	capacity=4, discipline=priority, mtbf=7200, mttr=300, on_failure=restart, process=normal(240,60)
unit_tests	machine	capacity=6, discipline=priority, process=normal(90,20)
int_env_1	machine	capacity=3, discipline=priority, mtbf=14400, mttr=900, process=normal(420,90)
int_env_2	machine	capacity=3, discipline=priority, mtbf=14400, mttr=900, process=normal(420,90)
int_env_3	machine	capacity=3, discipline=priority, mtbf=14400, mttr=900, process=normal(420,90)
deploy	machine	capacity=4, process=normal(90,20)
prod	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
Medium	**unit_tests**	Blocked 25% of the time: finished parts cannot leave because the next buffer is full.	-3
Low	**throughput**	Throughput 95% CI is +-5.3% of the mean.	-1
Low	**run length**	The measured window (7.5 h) is only 14x the cycle time.	-2
	Total deductions		**-6**
	Health score		**94**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.